

Learning to Reweight Examples for Robust Deep Learning

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Paper: Ren et al., ICML 2018

Code:

`https://github.com/lazarsalt14/DA5400-Project/tree/main/l2rw`

Motivation — Why Reweight Training Examples?

Class Imbalance

One class dominates the training distribution — common in:

- Fraud detection
- Medical diagnosis
- Rare-event prediction

Simple balancing methods break at extreme ratios (20:1).

Label Noise

Crowdsourced / weakly supervised labels are corrupted.

Neural networks:

- Memorise noisy labels
- Overfit wrong patterns
- Generalise poorly

[2]

Why L2RW?

Classical fixes (inverse weighting, resampling) are inconsistent when imbalance and noise co-exist.

L2RW:

- Learns per-example weights online
- Uses a clean validation set
- Requires no extra hyperparameters

[1]

L2RW Algorithm — Bi-Level Optimisation

Objective: Find per-example weights w^* such that using them minimises the *clean* validation loss.

Bi-level problem:

$$w^* = \arg\min_{w \geq 0} \frac{1}{M} \sum_j \mathcal{L}(\Phi(x_j^v; \theta^*(w)), y_j^v)$$

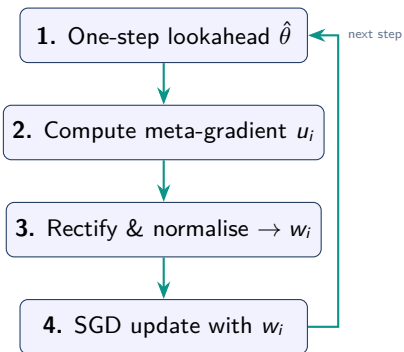
$$\theta^*(w) = \arg\min_{\theta} \sum_i w_i \mathcal{L}(\Phi(x_i; \theta), y_i)$$

Online approximation (one gradient step):

$$u_{i,t} \propto - \left. \frac{\partial \mathcal{L}_{\text{val}}(\hat{\theta}_{t+1})}{\partial \varepsilon_i} \right|_{\varepsilon=0}$$

Gradient similarity (Eq. 4):

$$u_{i,t} \propto - \frac{1}{m} \sum_j \sum_l \underbrace{(\tilde{z}_{j,l}^v \top \tilde{z}_{i,l})}_{\text{activation}} \underbrace{(g_{j,l}^v \top g_{i,l})}_{\text{gradient}}$$



Normalisation

$$\tilde{w}_i = \max(u_i, 0), \quad w_i = \frac{\tilde{w}_i}{\sum_j \tilde{w}_j + \delta}$$

Datasets — Two Synthetic Benchmarks

Dataset 1 — Credit (Class Imbalance)

Property	Value
Samples	2,000
Features	20 (10 informative)
Task	Binary (fraud)
Imbalance ratios	$r \in \{2, 5, 10, 15, 20\}:1$
Split	70/10/20%
Val & test	Stratified balanced

Dataset 2 — Phishing (Label Noise)

Property	Value
Samples	2,000
Features	15
Classes	4 (balanced)
Noise rates	$r \in \{0, 10, \dots, 50\}\%$
Noise types	Uniform flip, Bg flip
Val labels	Always clean (10%)

Suitability

Both datasets mirror the paper's experimental conditions: controlled imbalance/noise, clean validation splits, and fully CPU-reproducible experiments.

Experimental Setup & Baselines

L2RW Implementation

- 2-hidden-layer MLP: $64 \rightarrow 32$, ReLU, Softmax
- Pure **NumPy** (no autograd framework)
- He initialisation + `StandardScaler`
- $\alpha = 0.25$, 200 epochs, batch size 64
- Validation batch $m = |\mathcal{D}_g|$

Evaluation Metrics

- **F1 Macro** — primary imbalance metric
- **AUC-ROC**
- **G-Mean**
- Accuracy

Three Baselines

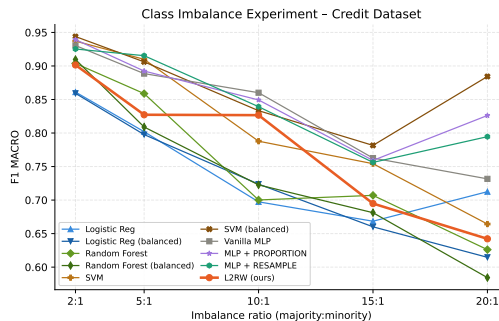
- 1 **Logistic Regression** — ℓ_2 regularised
- 2 **Random Forest** — 200 trees, Gini impurity
- 3 **SVM (RBF)** — balanced weighting

Additional MLP Variants

- **Vanilla MLP** — no reweighting
- **MLP + PROPORTION** — inverse-frequency loss
- **MLP + RESAMPLE** — balanced oversampling

Results — Class Imbalance (Credit Dataset, F1 Macro)

Model	2:1	5:1	10:1	15:1	20:1
Logistic Reg	.861	.802	.697	.668	.713
LR (balanced)	.861	.808	.707	.660	.615
Random Forest	.904	.859	.700	.707	.626
RF (balanced)	.898	.856	.710	.681	.584
SVM	.937	.910	.788	.754	.664
SVM (bal.)	.937	.936	.839	.782	.884
Vanilla MLP	.930	.889	.860	.763	.732
MLP+PROP	.939	.892	.849	.759	.826
MLP+RESAMP	.925	.915	.839	.756	.795
L2RW (ours)	.902	.827	.827	.695	.642

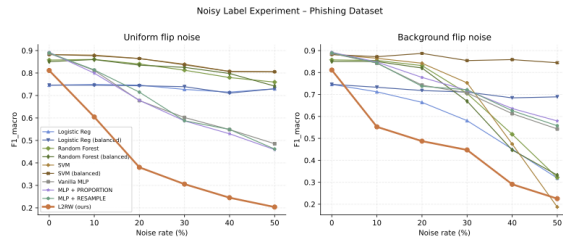
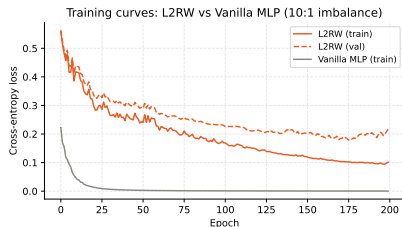


Key Takeaway

SVM (balanced) dominates at extreme imbalance ($F1 = 0.884$ at 20:1). L2RW remains competitive at moderate ratios, but performance drops when the minority validation set becomes extremely small.

Results — Label Noise (Phishing Dataset, Accuracy)

Model	0%	10%	20%	30%	40%	50%
Logistic Reg	.748	.750	.748	.728	.718	.730
Rand.Forest	.858	.863	.843	.812	.785	.765
SVM	.882	.880	.865	.840	.807	.807
Vanilla MLP	.887	.812	.677	.605	.550	.490
L2RW (ours)	.812	.627	.445	.340	.245	.207



Discussion, Limitations & Conclusion

Confirmed Claims

- Near-zero weights on corrupted examples
- Meta-weights reduce over-fitting
- No extra hyperparameters vs SGD

Quantitative Gaps

- Tabular MLP vs CIFAR ResNet
- Small-scale setup ($N = 2000$)
- Tiny minority validation set

Limitations

- Meta-gradients $\sim 3\times$ slower
- Requires unbiased validation set
- Early ReLU weight collapse

Key Finding

- Balanced SVM outperforms L2RW
- Tree/margin methods remain strong
- Consistent with Grinsztajn et al. (2022)

Conclusion: L2RW successfully identifies noisy samples, but on structured tabular data, classical margin-based methods remain more effective and efficient.

References I



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